

Product Management Technical Report (Revised)

Executive Summary

Laravel Product Management Application with authentication, authorization, CRUD operations, search, filtering, pagination, Docker deployment, Redis integration, CI quality gates, and layered architecture.

System Architecture

Layered architecture using Routes, Middleware, Controllers, Form Requests, Services, Repositories, Models, Database, Redis, and Blade Views.

Architecture Decisions

Repository Pattern, Service Layer, Form Request Validation, Policies, Authorization, and Dependency Injection.

Authentication and Authorization

Laravel Breeze authentication, email verification, role-based authorization, policies, middleware, password security, session security, and rate limiting.

Product Management Module

Product CRUD, SKU generation, stock status derivation, search, filtering, sorting, pagination, rich text editor support, and validation.

Challenges and Solutions

Controller separation, authorization, Docker automation, rich text sanitization, search scalability, and testing strategy.

Security and Performance

SQL injection protection, XSS prevention, CSRF protection, mass assignment protection, authentication security, Redis caching, indexing, and performance considerations.

Testing Strategy

Feature and unit testing with Pest/PHPUnit covering CRUD, validation, inventory, authorization, search, and filtering.

Deployment Strategy

Docker Compose architecture with PHP-FPM, Nginx, MariaDB, Redis, Queue Worker, Scheduler, and automated bootstrap process.

Code Quality Compliance

PSR-12 coding standards, dependency injection, service/repository patterns, policies, middleware, validation, and security compliance.

CI/CD Roadmap

GitHub Actions pipeline including quality checks, tests, audits, Docker build, deployment approvals, health checks, and rollback planning.

Future Improvements

• Product Image Upload with secure validation and storage. • Soft Deletes with Restore functionality. • Bulk Delete and Bulk Update operations. • Export CSV and Excel reports. • Created By User and Updated By User tracking. • Product Activity Log and Audit Trail. • Laravel Scout integration using Meilisearch/Elasticsearch for large datasets. • Advanced Search and Filtering. • Checkout and Payment Gateway Integration. • Invoice and Receipt PDF Generation. • Order Tracking and Order Status Timeline. • Customer Dashboard. • Admin Dashboard with Analytics. • Product Categories and Inventory Alerts. • Sentry Error Monitoring and Application Tracking. • MFA for administrator accounts. • OAuth / Social Login support. • Kubernetes and Blue-Green deployment support. • OpenTelemetry observability and structured logging.

Final Assessment

Strong architecture, maintainability, security, testing foundation, Docker deployment, and scalability improvements planned for future releases.